MOS · A0 RESEARCH POSTER

KNOWLEDGE IS GEOMETRY · REASONING IS DYNAMICS · CONSISTENCY IS COHOMOLOGY

THE MATHEMATICS OF THE MATHEMATICAL OPERATING SYSTEM

A Cellular–Sheaf Architecture for Structured Machine Cognition

$$M = (C, \mathcal{F}, s)$$

M

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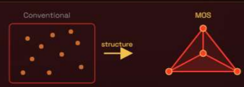
MEMORY > STRUCTURE > CONSISTENCY > GEOMETRY > GROWTH > COMPUTATION > VERIFICATION > FUTURE

01 MEMORY · MOTIVATION

Why memory as geometry?

Memory $\longrightarrow$ Structured Object



Embeddings retrieve well but are blind to how concepts relate. MOS makes relationships **intrinsic to the space.**

02 STRUCTURE · STATE SPACE

Cognition is a single triple

$$M = (C, \mathcal{F}, s)$$

$s \in H^0(C, \mathcal{F})$

Complex **C**, sheaf $\mathcal{F}$, coherent global section **s.**

03 STRUCTURE · BASE COMPLEX

Beyond graphs

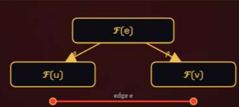
$$C = (V, \Sigma)$$

A triangle is **one** relation among three concepts — not three edges — so MOS carries real topology.

04 CONSISTENCY · CELLULAR SHEAF

Attaching local information

$$\mathcal{F} : C \longrightarrow \mathbf{Vect}$$

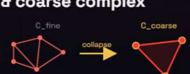


$\rho_{\tau\to\sigma} : \mathcal{F}(\tau) \to \mathcal{F}(\sigma)$

Stalks + restriction maps ρ; compatibility is *enforced by structure.*

L DETAIL · TWO SCALES

Fine complex & coarse complex



δC_{fine} δC_{coarse}

The **fine** complex holds every raw concept and relation; the **coarse** complex is its quotient into stable coalitions. Growth acts on both — fine detail feeds coarse abstraction; coarse structure constrains fine drift.

05 CONSISTENCY · COHERENCE

Consistency, made a number

$$L = \delta^{\mathsf{T}}\delta \qquad \omega(x) = \|\delta x\|^2$$

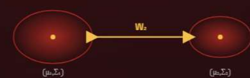
$\omega(x) = x^{\mathsf{T}} L x, \quad \rho = \frac{\omega(x)}{B\|x\|^2} \in [0,1]$

Disagreement energy vanishes exactly on global sections: **H^0=ker L.**

06 GEOMETRY · SEMANTIC GEOMETRY

Concepts as clouds, not points

$$X_i \sim \mathcal{N}(\mu_i, \Sigma_i)$$



$\mu = (\sum_i \Sigma_i^{-1})^{-1}(\sum_i \Sigma_i^{-1}\mu_i)$

Similarity is Wasserstein W_2; evidence fuses by precision-weighting.

07 DIAGNOSTICS · LOCALIZING

Not *whether* — but *where*

$$b_0 \qquad H^1(C, \mathcal{F})$$

Measure → Locate → Analyze → Repair

Invariants **b_0, H^1** separate local conflict from global obstruction.

08 GROWTH · DYNAMICS

Learning is a change in shape

$$\mathcal{D}(t) \longrightarrow \mathcal{D}(t+1)$$

$\delta C_{\text{fine}},\ \delta C_{\text{coarse}},\ \delta\mathcal{F},\ \delta\mathcal{D} \quad w_{t+1} = (1-\gamma)w_t$

Four growth axes evolve at once — concepts, coalitions, sheaf data, and the operator algebra $\mathcal{D}$ itself. Only **VerifyOp**-passing procedures become new operators.

L DETAIL · EMERGENT OPERATORS

Microneurons → operator algebra

persists $\Rightarrow$ new generator of $\mathcal{D}$

A recurring reasoning subgraph — a **microneuron** — that survives verification is promoted to a first-class operator. The algebra $\mathcal{D}$ is learned, not fixed: the system writes its own vocabulary.

09 COMPUTATION · OPERADS

Reasoning as composition

$$T = (N, \prec)$$

$L_0, L_1, L_2, \ldots$ (antichains)

A DAG of operators; antichain layers run in **parallel.**

10 COMPUTATION · LIFT–PROJECT

Language ↔ computation

π : Lang $\to$ DAG $\quad G$: DAG $\to$ Lang

$\ker(\pi)$ = semantically void input

Void input lands in **ker(π).** L-⊣M is a *guiding principle, not yet proved.*

11 VERIFICATION · EXTERNAL ORACLE

Coherence is not truth

Coherence $\neq$ Truth

exit $\in \{0,1,2\} \quad \Omega \leftarrow \Omega + 1000$

Absence of proof is not proof of absence: unverifiable ≠ refuted.

12 VERIFICATION · CONTROLLER

One number gates the system

$$\text{mode}(x) = \begin{cases} \textbf{Resolve} & \rho > \varepsilon \\ \textbf{Explore} & \rho \le \varepsilon \\ \textbf{Unknown} & \rho = \perp \end{cases}$$

$\varepsilon \approx 0.10$

Above ε → **Resolve**; below → **Explore**; no measurement → Unknown (never Explore).

L DETAIL · PROVENANCE

Every state carries its history

Each mutation — bind, split, merge, verify — is logged as an ordered **provenance trail.** A state is never anonymous: it knows which operators built it, so conflicts trace to their origin and roll back.

13 FUTURE · CLAIMS, NON-CLAIMS & INVITATION

IS claimed

- ✓ measurable coherence
- ✓ structured memory
- ✓ verifiable growth
- ✓ mathematical substrate

NOT claimed

- ✗ intelligence proven
- ✗ AGI
- ✗ universal reasoning

Toward constitutive reasoning

1. Sheaf diffusion as inference $\dot{x} = -Lx$
2. Non-trivial *learned* restriction maps
3. Higher-cell Laplacians
4. Persistence-driven structural learning

Status & an invitation

MOS is an **early-stage research program** — years from its final form; the definitions are laid, but many components are still under active development. This poster is deliberately an **invitation:** I am presenting it to gather criticism, suggestions and collaboration from the viewers and the jury. Tell me where it breaks.